DATA ANALYTICS — PROJECT 2

Exploratory Data Analysis (EDA) | Industrial Training Submission

Objective: Analyze the dataset to understand patterns, trends, distributions, relationships, and potential outliers.

1. Dataset Overview

The analysis uses **1,200 records** and **14 columns**. Date coverage: **2023-01-01 to 2025-06-30**. The workbook contains the data used for analysis, descriptive statistics, categorical counts, monthly trends, product and city summaries, outlier audit, and correlation results.

2. Descriptive Statistics

Metric	Count	Mean	Median	Minimum	Maximum
Quantity	1200	2.95	3.00	1.00	5.00
ItemsInCart	1200	5.48	5.00	1.00	10.00
UnitPrice	1200	356.41	364.21	11.39	699.93
TotalPrice	1200	1053.97	823.62	11.39	3456.40

3. Key Observations

• The dataset contains 1,200 records and 14 columns, covering 2023-01-01 to 2025-06-30.

• Total revenue across the dataset is 1,264,761.96; average transaction value is 1,053.97 and median transaction value is 823.62.

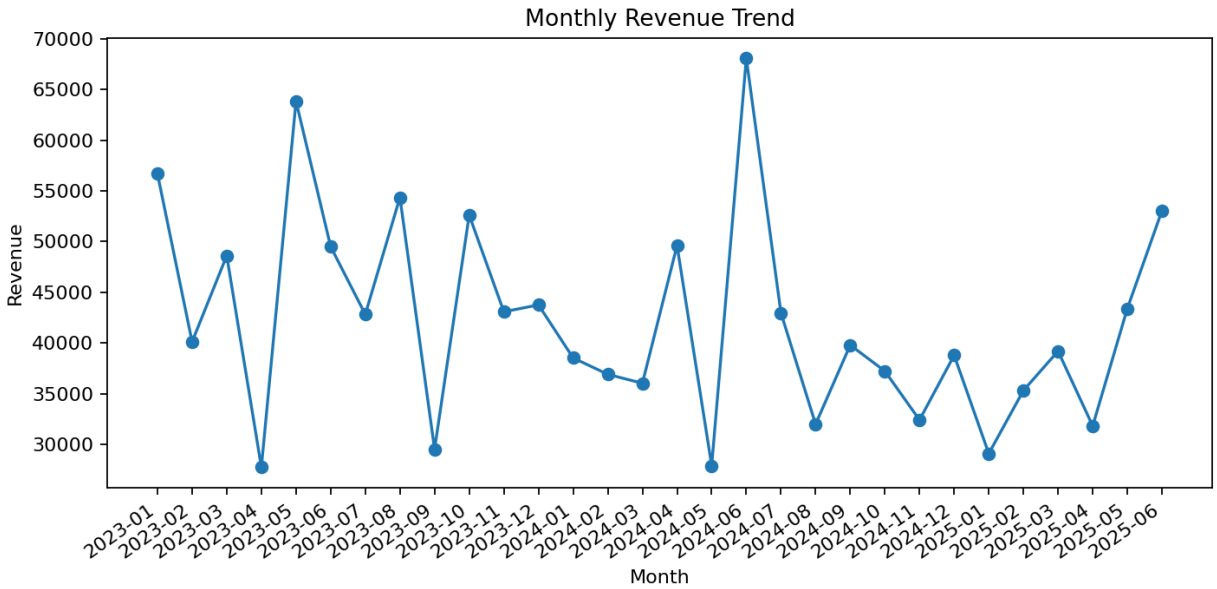
• Highest revenue product is Chair with revenue of 195,620.11.

• Most frequent order status is Cancelled (250 records, 20.8%).

• Highest-revenue month is 2024-06 (68,068.54 revenue).

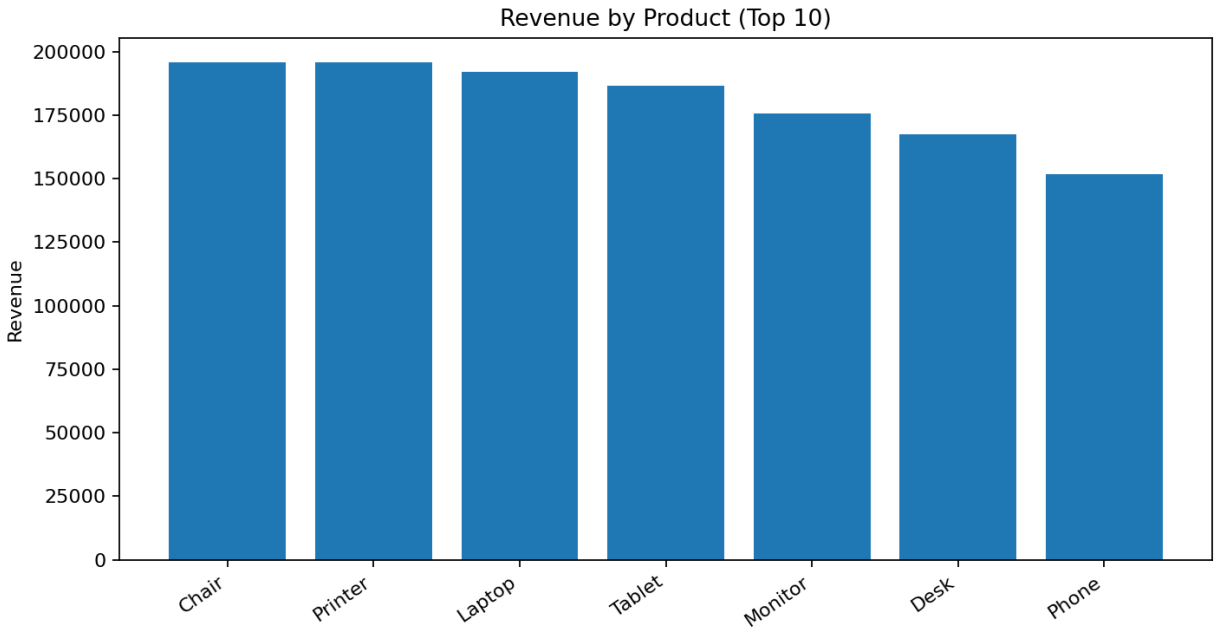
• Potential outliers were flagged using the 1.5×IQR rule. These are candidates for investigation, not automatic errors; unusual transactions should not be deleted without business context.

4. Trend Analysis

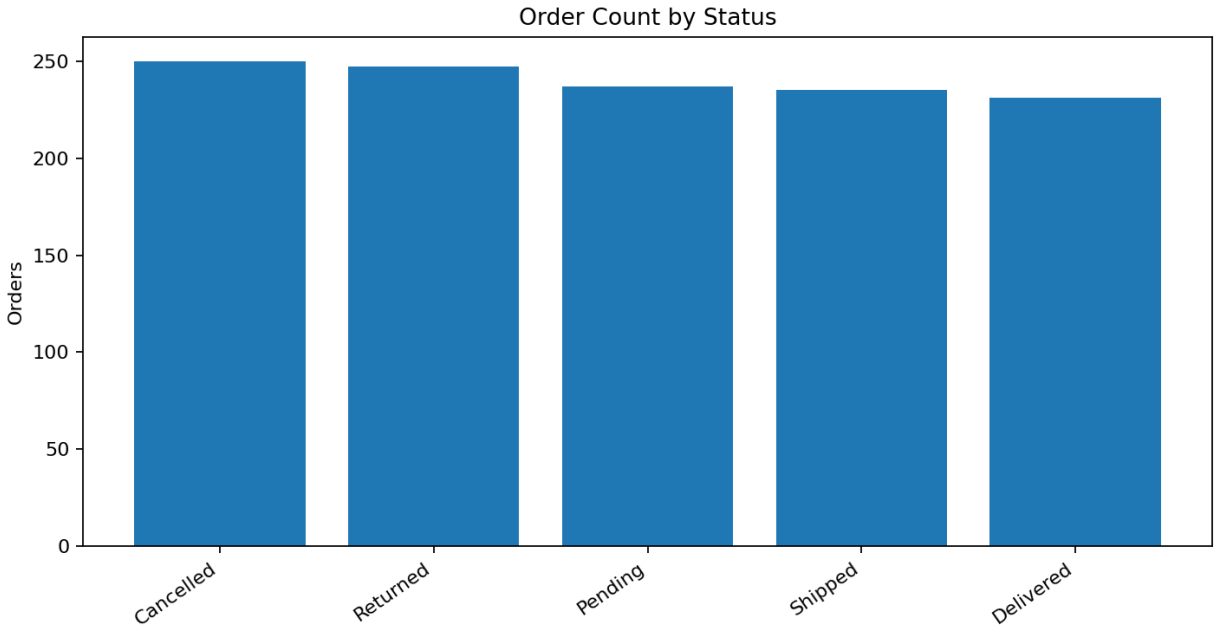


The monthly revenue chart helps compare revenue over time. Interpret peaks and dips in context; the chart describes association over time and does not by itself explain causes.

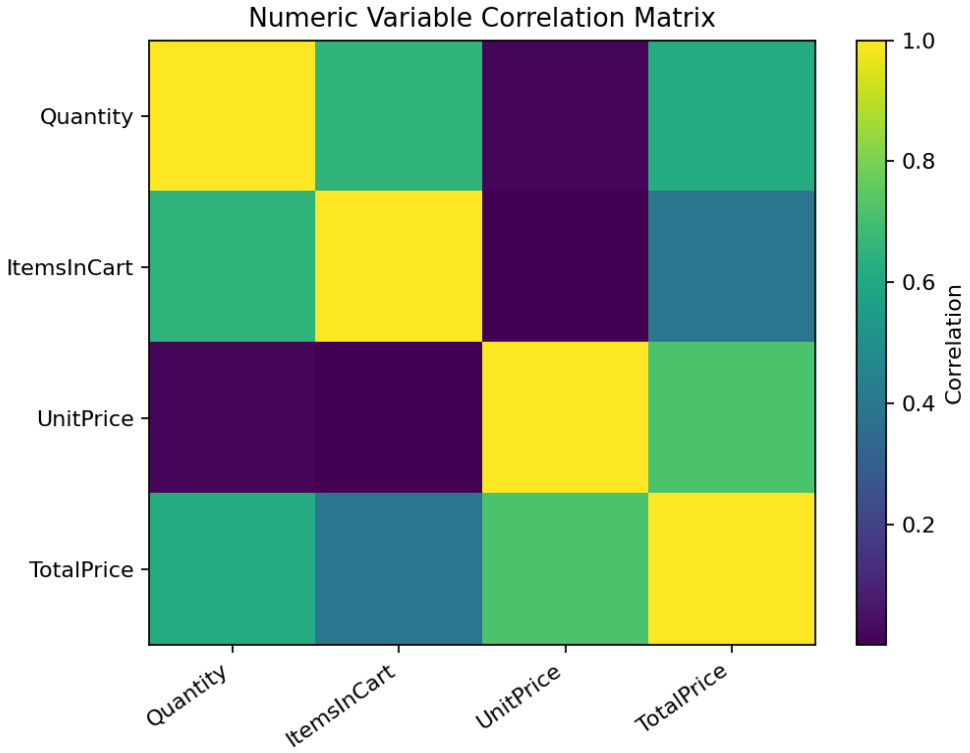
5. Product Performance



7. Order Status Distribution



8. Correlation Analysis



Correlation measures the strength and direction of linear association between numeric variables. Correlation does not prove causation; hidden variables and business context should be considered.

Outlier Method

Potential outliers were identified using the Interquartile Range (IQR) rule: values below Q1 – 1.5×IQR or above Q3 + 1.5×IQR are flagged. An outlier is not automatically an error. Investigate flagged values against the source and business context before changing or removing them.

Conclusion

The EDA summarizes the dataset using descriptive statistics, category counts, time-based trends, product and city comparisons, correlation analysis, and an IQR-based outlier audit. Detailed results are available in the accompanying Excel workbook. These findings provide a starting point for business decisions and further analysis; correlation and outlier flags should be interpreted with context.